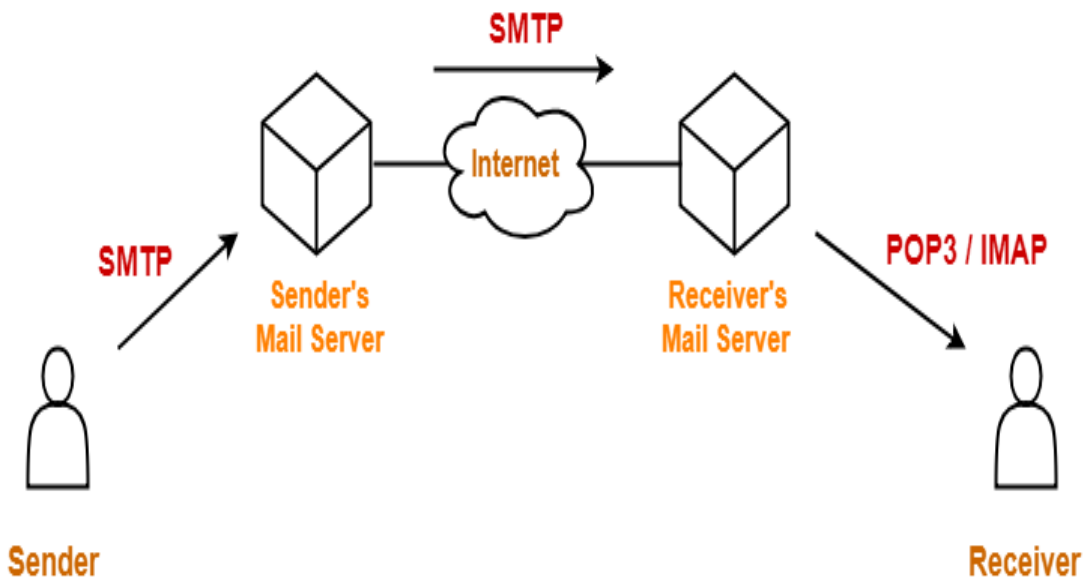
Simple Mail Transfer Protocol-

- SMTP is short for **Simple Mail Transfer Protocol.**
- It is an application layer protocol.
- It is used for sending the emails efficiently and reliably over the internet.

Working-

- SMTP server is always on a listening mode.
- Client initiates a TCP connection with the SMTP server.
- SMTP server listens for a connection and initiates a connection on that port.
- The connection is established.
- Client informs the SMTP server that it would like to send a mail.
- Assuming the server is OK, client sends the mail to its mail server.
- Client's mail server use DNS to get the IP Address of receiver's mail server.
- Then, SMTP transfers the mail from sender's mail server to the receiver's mail server.



While sending the mail, SMTP is used two times-

1. Between the sender and the sender's mail server
2. Between the sender's mail server and the receiver's mail server

NOTE-

To receive or download the email,

- Another protocol is needed between the receiver's mail server and the receiver.
- The most commonly used protocols are POP3 and IMAP.

Characteristics of SMTP-

- SMTP is a push protocol.
- SMTP uses TCP at the transport layer.
- SMTP uses port number 25.
- SMTP uses persistent TCP connections, so it can send multiple emails at once.
- SMTP is a connection oriented protocol.
- SMTP is an in-band protocol.
- SMTP is a stateless protocol.

Important Points-

Note-01:

SMTP is a pure text based protocol.

- SMTP can only handle the messages containing 7 bit ASCII text.
- SMTP cannot transfer other types of data like images, video, audio etc.
- SMTP cannot transfer executable files and binary objects.
- SMTP cannot transfer the text data of other languages like French, Japanese, Chinese etc.

(since they are represented in 8 bit codes)

Note-02:

MIME extends the limited capabilities of email.

As the name suggests,

- Multipurpose Internet Email Extension (MIME) is an extension to the internet email protocol.
- It extends the limited capabilities of email by enabling the users to send and receive graphics, audio files, video files etc in the message.
- MIME was specially designed for SMTP.

Note-03:

SMTP is a stateless protocol.

This is because-

- It does not maintain the state of its clients.
- If an email is asked to be sent twice, then SMTP server resends it without saying that the email has already been sent

Note-04:

We cannot use SMTP at the receiver's side.

This is because-

- SMTP is a push protocol.
- At receiver's side, a pull protocol like POP3, IMAP is needed.
- Receiver periodically checks if he has any mail from his mail server.

Note-05:

Sender and receiver cannot run SMTP between their machines.

This is because-

- Machines cannot always be ON.
- So, the functionality has been divided between the client and the mail server.

- The mail server receives the mail on behalf of its client and manages the mail box of the client.

Note-06:

SMTP is not suitable for client authentication.

This is because-

- SMTP does not require authentication.
- It allows anyone on the Internet to send emails to anyone or even to a large group of people.
- SMTP Auth short for SMTP Authentication has been provided for authentication.

Post Office Protocol-

- POP is short for **Post Office Protocol**.
- It is an application layer protocol.

Purpose-

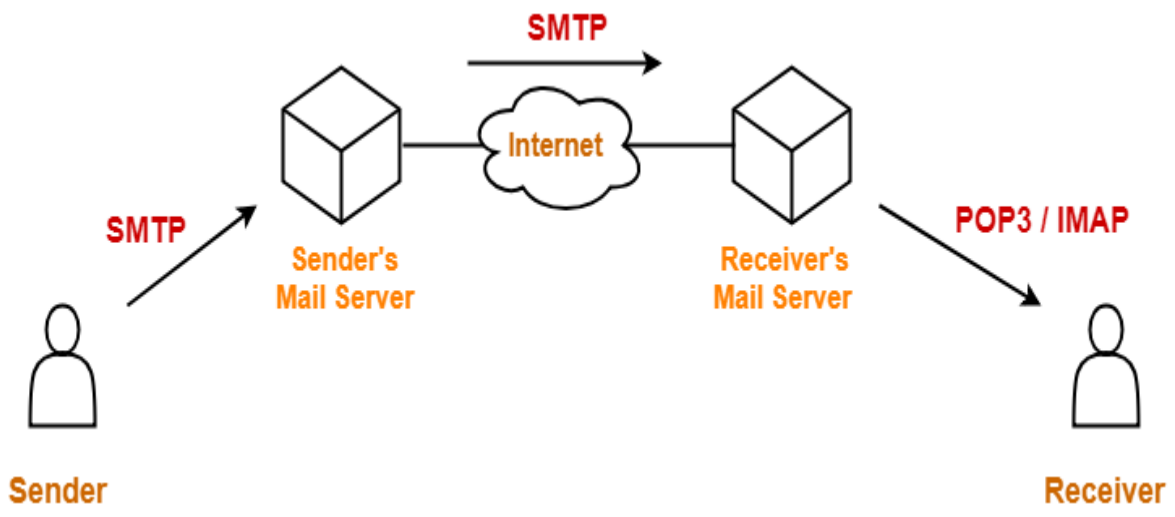
- It is a message access protocol.
- It enables the clients to receive or download the emails from their remote mail server.
- POP version 3 (POP3) is the most popularly used version.

Working-

To retrieve a message from the mail server-

- Client establishes a TCP connection using port 110.
- Client identifies itself to the server.

- Client issues a series of POP3 commands.



Characteristics of POP-

- POP is a pull protocol.
- POP uses TCP at the transport layer.
- POP uses port number 110.
- POP uses persistent TCP connections.
- POP is a connection oriented protocol.
- POP is an in-band protocol.
- POP is a stateful protocol until the mail is downloaded as well as stateless across sessions.

Internet Message Access Protocol-

- IMAP is short for **Internet Message Access Protocol**.
- It is an application layer protocol.
- It also enables the clients to receive or download the emails from their remote mail server.

POP has been largely superseded by Internet Message Access Protocol (IMAP)

Characteristics of IMAP-

- IMAP is a pull protocol.
- IMAP uses TCP at the transport layer.
- IMAP uses port number 143.
- IMAP uses persistent TCP connections.
- IMAP is a connection oriented protocol.
- IMAP is an in-band protocol.
- IMAP is a stateful protocol.
- IMAP distributes mail boxes across multiple servers.

POP Vs IMAP-

The following table lists the differences between POP and IMAP-

Post Office Protocol	**Internet Message Access Protocol**
POP allows you to read the mail only after downloading it.	IMAP allows you to check the mail content before downloading. So, with IMAP you can choose to download your messages or just delete them.
The email message is downloaded to the local computer and the copy at the server is deleted.	The email message is stored on the mail server itself.
The user can not organize mails in the mail box of the mail server.	The user can organize mails on the server.

Mails can only be accessed from a single device.	Mails can be accessed from multiple devices which is the biggest advantage.
In POP, the mail server and client's mail account are not synced. So, changes made in the client's mail account are not visible on the web mail inbox. <u>Example-</u> If you are using POP and marks a mail as read, it does not get marked as read in the web mail inbox because the mails are downloaded to the computer and so the changes won't be visible on the server.	In IMAP, the mail server and the client's mail account are synced. So, changes made in the client's mail account are instantly visible on the web mail inbox. <u>Example-</u> If you are using IMAP and marks a mail as read, it gets marked as read in the web mail inbox too because the changes are taking place on the server.
POP is a stateful protocol until the mail is downloaded as well as stateless across sessions.	IMAP is a stateful protocol because the IMAP server has to maintain a folder hierarchy for each of its users.
POP is a better choice for those who hardly checks their mail on any other computer.	IMAP is a better choice for those who frequently check their mails on other computers.

<u>Hyper Text Transfer Protocol-</u>

- HTTP is short for **Hyper Text Transfer Protocol**.
- It is an application layer protocol.

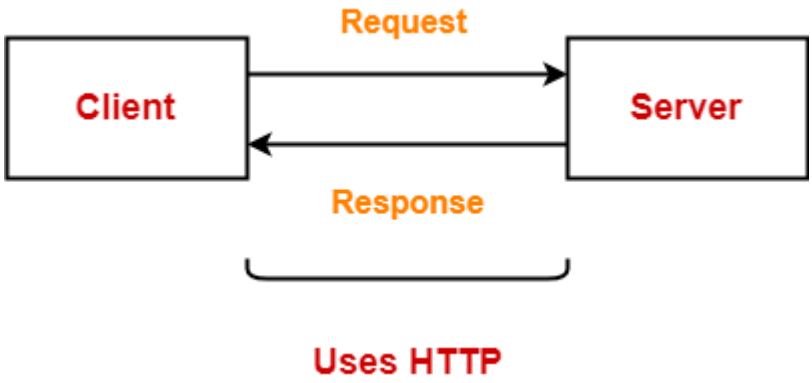
<u>Purpose-</u>

- It is mainly used for the retrieval of data from websites throughout the internet.
- It works on the top of TCP/IP suite of protocols.

Working-

HTTP uses a client-server model where-

- Web browser is the client.
- Client communicates with the web server hosting the website.



Whenever a client requests some information (say clicks on a hyperlink) to the website server.

The browser sends a request message to the HTTP server for the requested objects.

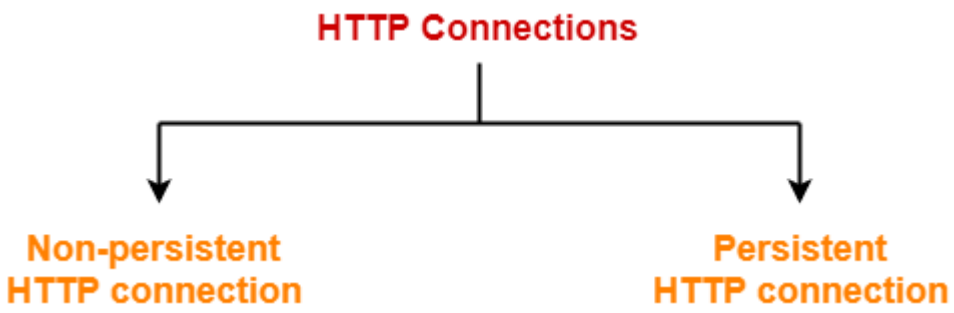
Then-

- HTTP opens a connection between the client and server through **TCP**.
- HTTP sends a request to the server which collects the requested data.
- HTTP sends the response with the objects back to the client.
- HTTP closes the connection.

HTTP Connections-

HTTP connections can be of two types-

1. Non-persistent HTTP connection
2. Persistent HTTP connection



Non-persistent HTTP connection	**Persistent HTTP connection**
Non-persistent HTTP connection is one that is used for serving exactly one request and sending one response.	Persistent HTTP connection is one that can be used for serving multiple requests.
HTTP server closes the TCP connection automatically after sending a HTTP response.	HTTP server closes the TCP connection only when it is not used for a certain configurable amount of time.
A new separate TCP connection is used for each object.	A single TCP connection is used for sending multiple objects one after the other.
HTTP 1.0 supports non-persistent connections by default.	HTTP 1.1 supports persistent connections by default.

Example-	Example-
Suppose a request has been made for a HTML page that contains 10 images (called objects). Then, With non-persistent connection, all the 11 objects (1 page + 10 images) will be sent one by one. For getting each object, a new separate connection will be opened and used.	Suppose a request has been made for a HTML page that contains 10 images (called objects). Then, With persistent connection, all the 11 objects (1 page + 10 images) will be sent one after the other using a single TCP connection.

Important Notes-

Note-01:

HTTP uses TCP at the transport layer

This is because-

- Unlike **UDP**, it guarantees the delivery of data via a **Three-way handshake**.
- It ensures the re transmission of lost packets.
- HTTP does not have any inbuilt facility for providing reliability.
- So, if HTTP uses UDP, then it will have to maintain or handle the session on its own.
- For example- If a packet gets lost, then HTTP will have to re-transmit the packet.

Note-02:

It is important to know-

- Any service which does not use TCP should have the inbuilt facility for providing reliability.

Note-03:

HTTP uses port number 80.

- HTTP clients uses port 80 to send and receive requested web pages from a HTTP server.
- Similarly, HTTP server responds to all the requests at port 80.

Note-04:

HTTP 1.0 is non-persistent and HTTP 1.1 is persistent

- Already discussed in the above table.
- Persistent connections improve the performance by 20%

Note-05:

HTTP 1.0 is a connectionless protocol.

This is because-

- After serving the single HTTP request, the connection is closed and it is not used again.
- So, HTTP 1.0 without connection keep alive is connectionless.

Note-06:

HTTP is an in-band protocol

This is because-

- HTTP passes the control data (commands) and main data over the same connection.
- Both control data and main data are processed in the same way without any distinction.
- No high priority is given to the control data (commands)

Note-07:

HTTP is a stateless protocol

This is because-

- HTTP server does not maintain any state.

- It forgets about the client after sending the response.
- It treats every new request independently.
- HTTP closes the connection automatically after generating the response for each request.
- This ensures that no client can engage connection with web server for a long time

What If HTTP Is Stateful Protocol?

If HTTP is a stateful protocol, then-

- It will give a chance to the browser window to engage the connection with the web server for a long time.
- This may unnecessarily create a situation of reaching to maximum connections of a web server even though most of the connections are idle.

File Transfer Protocol-

- FTP is short for **File Transfer Protocol**.
- It is an application layer protocol.

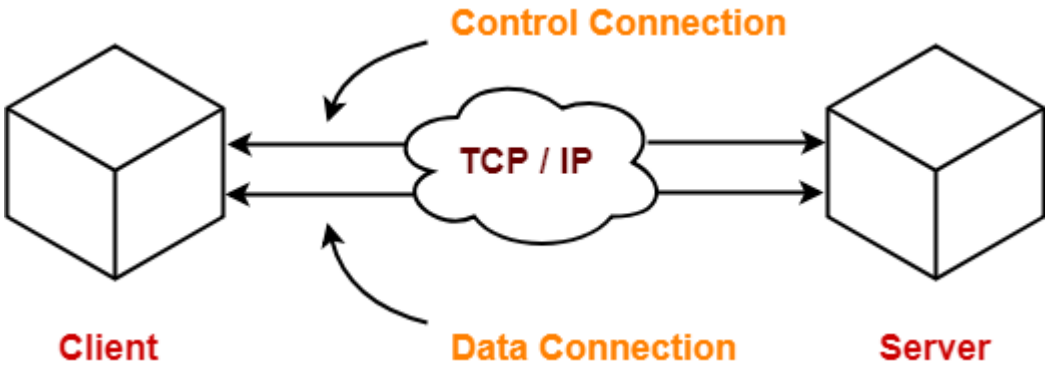
Purpose-

- It is used for exchanging files over the internet.
- It enables the users to upload and download the files from the internet.

Working-

FTP establishes two TCP connections between the client and the server.

- One connection is used for transferring data.
- Other connection is used for transferring control information.



Characteristics of FTP-

- FTP uses TCP at the transport layer.
- FTP uses port number 21 for control connection.
- FTP uses port number 20 for data connection.
- FTP uses persistent TCP connections for control connection.
- FTP uses non-persistent connections for data connection.
- FTP is a connection oriented protocol.
- FTP is an out-of-band protocol as data and control information flow over different connections.
- SMTP is a stateful protocol.

Important Notes-

Note-01:

Emails can't be sent using FTP.

This is because-

- FTP requires the connection establishment between the client and server before transferring the files.
- So, both have to be online at the same time.

- That is why, emails are not sent using FTP.

Note-02:

FTP can transfer one file at a time

- FTP is used for transferring one file at a time in either direction between the client and the server.

Note-03:

FTP is a stateful protocol.

This is because-

- The client establishes control connection for the duration of an FTP session.
- It typically spans multiple data transfers.
- So, FTP is a stateful protocol.

DNS in Networking-

- DNS is short for **Domain Name Service** or **Domain Name System**.
- It is an application layer protocol.

Purpose-

- DNS is a host name to IP Address translation service.
- It converts the names we type in our web browser address bar to the IP Address of web servers hosting those sites.

Need-

The need for Domain Name Service arises due to the following reasons-

Point-01:

- **IP Addresses** are not static and may change dynamically.
- So, a mapping is required which maps the domain names to the IP Addresses of their web servers.

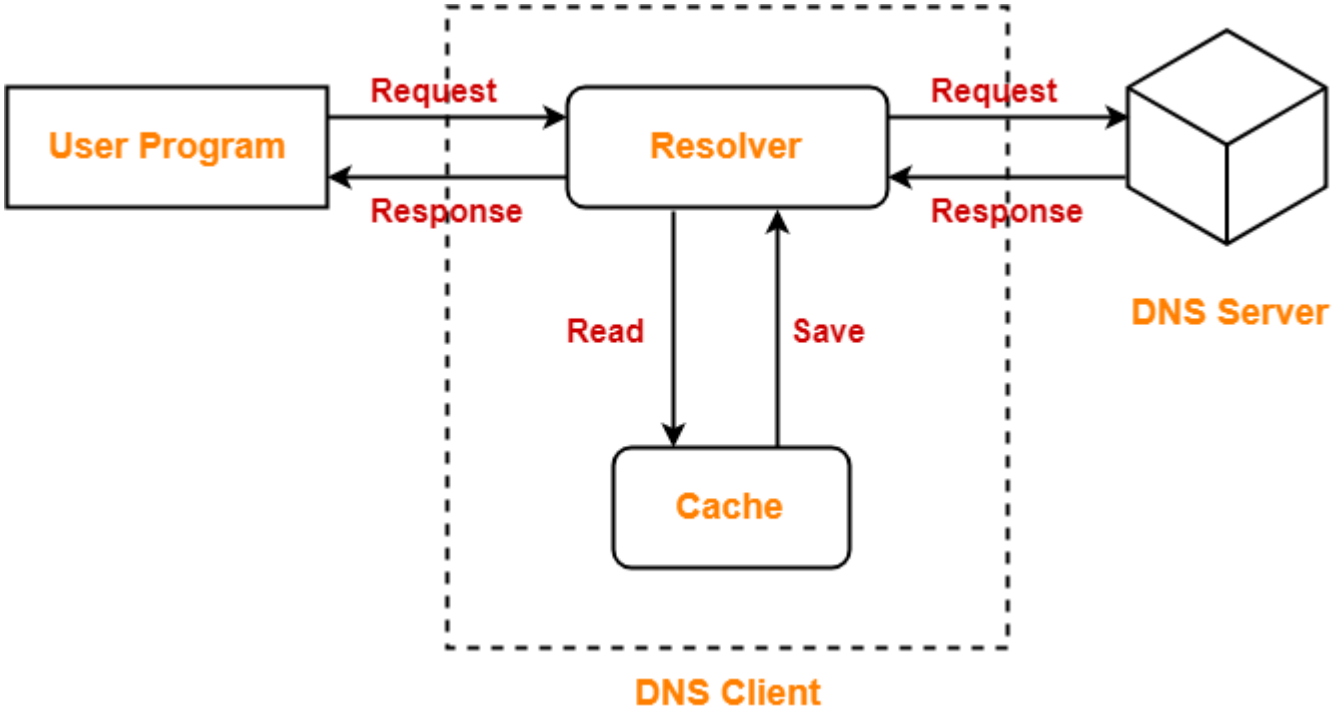
Point-02:

- IP Addresses are a complex series of numbers.
- So, it is difficult to remember IP Addresses directly while it is easy to remember names.

DNS Resolution-

DNS Resolution is a process of resolving a domain name onto an IP Address.

The following diagram illustrates the process of DNS resolution-



The steps involved in DNS Resolution are-

Step-01:

- A user program sends a name query to a library procedure called the resolver.

Step-02:

Resolver looks up the local domain name cache for a match.

- If a match is found, it sends the corresponding IP Address back.
- If no match is found, it sends a query to the local DNS server.

Step-03:

DNS server looks up the name.

- If a match is found, it returns the corresponding IP Address to the resolver.
- If no match is found, the local DNS server sends a query to a higher level DNS server.
- This process is continued until a result is returned.

Step-04:

- After receiving a response, the DNS client returns the resolution result to the application.

Important Notes-

Note-01:

DNS uses UDP (port 53) at the transport layer.

DNS uses **UDP** at the transport layer due to the following reasons-

Point-01:

- UDP is much faster than **TCP**.
- TCP is slow as it uses **Three-way handshake** to start the data transfer.

Point-02:

- DNS requests are very small.
- So, they fits well within UDP segments.

Point-03:

- Although UDP is not reliable but reliability can be added on application layer.
- Reliability can be added by using timeouts and resend at the application layer.

Thus, in the end both speed and protection are achieved.

Note-02:

DNS is a connection less protocol.

- DNS uses UDP at the transport layer for replying to the DNS queries of clients.
- Therefore, it is a connection less protocol

Note-03:

DNS is non-persistent.

Note-04:

DNS is a stateless protocol.

This is because-

- DNS server accepts the requests, process them, resolves the query and forget about them.
- It does not make any assumption how long this will be.

Note-05:

Mapping an IP Address onto a domain name is referred to as **Inverse domain**.

It is important to note-

- DNS can translate a domain name onto an IP Address.
- Also, it can translate an IP Address onto a domain name.

Note-06:

For the first time,There is more delay in translating the domain name onto an IP Address.

- Converting a domain name onto an IP Address is an extra overhead.
- This overhead is called as **DNS Overhead**.

 It causes an unnecessary delay in serving the request.

 So, there is more delay for the first time.

 To reduce the delay next time, IP Addresses are stored in the computer using log.

 This avoids the DNS overhead next time and takes less time in serving the request.

 When it gets expired, the request is again served through DNS.

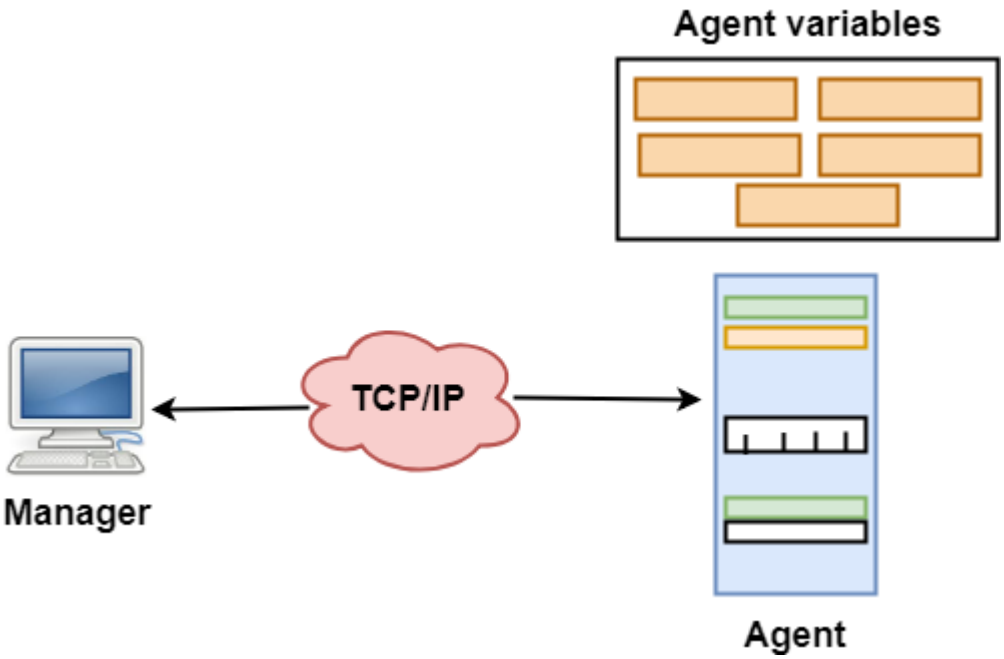
Simple Network Management Protocol (SNMP) –

If an organization has 1000 devices then to check all devices, one by one every day, are working properly or not is a hectic task. To ease these up, Simple Network Management Protocol (SNMP) is used.

Simple Network Management Protocol (SNMP) –

SNMP is an application layer protocol that uses UDP port number 161/162.SNMP is used to monitor the network, detect network faults, and sometimes even used to configure remote devices.

SNMP Concept



- SNMP has two components Manager and agent.
- The manager is a host that controls and monitors a set of agents such as routers.
- It is an application layer protocol in which a few manager stations can handle a set of agents.

- The protocol designed at the application level can monitor the devices made by different manufacturers and installed on different physical networks.
- It is used in a heterogeneous network made of different LANs and WANs connected by routers or gateways.

SNMP components –

There are 3 components of SNMP:

1. **SNMP Manager –**
   It is a centralized system used to monitor network. It is also known as Network Management Station (NMS)

2. **SNMP agent –**
   It is a software management software module installed on a managed device. Managed devices can be network devices like PC, routers, switches, servers, etc.

3. **Management Information Base –**
   MIB consists of information on resources that are to be managed. This information is organized hierarchically. It consists of objects instances which are essentially variables.

SNMP messages –

Different variables are:

1. **GetRequest –**
   SNMP manager sends this message to request data from the SNMP agent. It is simply used to retrieve data from SNMP agents. In response to this, the SNMP agent responds with the requested value through a response message.

2. **GetNextRequest –**
   This message can be sent to discover what data is available on an SNMP agent. The SNMP manager can request data continuously until no more data is left. In this way, the SNMP manager can take knowledge of all the available data on SNMP agents.

3. **GetBulkRequest –**
   This message is used to retrieve large data at once by the SNMP manager from the SNMP agent. It is introduced in SNMPv2c.

4. **SetRequest –**
   It is used by the SNMP manager to set the value of an object instance on the SNMP agent.

5. **Response –**
   It is a message sent from the agent upon a request from the manager. When sent in response to Get messages, it will contain the data requested. When sent in response to the Set message, it will contain the newly set value as confirmation that the value has been set.

6. **Trap –**
   These are the message sent by the agent without being requested by the manager. It is sent when a fault has occurred.

7. **InformRequest –**
   It was introduced in SNMPv2c, used to identify if the trap message has been received by the manager or not. The agents can be configured to send trap message continuously until it receives an Inform message. It is the same as a trap but adds an acknowledgement that the trap doesn't provide.

SNMP security levels –

It defines the type of security algorithm performed on SNMP packets. These are used in only SNMPv3. There are 3 security levels namely:

1. **noAuthNoPriv –**
   This (no authentication, no privacy) security level uses a community string for authentication and no encryption for privacy.

2. **authNopriv –** This security level (authentication, no privacy) uses HMAC with Md5 for authentication and no encryption is used for privacy.

3. **authPriv –** This security level (authentication, privacy) uses HMAC with Md5 or SHA for authentication and encryption uses the DES-56 algorithm.

SNMP versions –

There are 3 versions of SNMP:

1. **SNMPv1 –**
   It uses community strings for authentication and uses UDP only.

2. **SNMPv2c –**
   It uses community strings for authentication. It uses UDP but can be configured to use TCP.

3. **SNMPv3 –**
   It uses Hash-based MAC with MD5 or SHA for authentication and DES-56 for privacy. This version uses TCP. Therefore, the conclusion is the higher the version of SNMP, the more secure it will be.

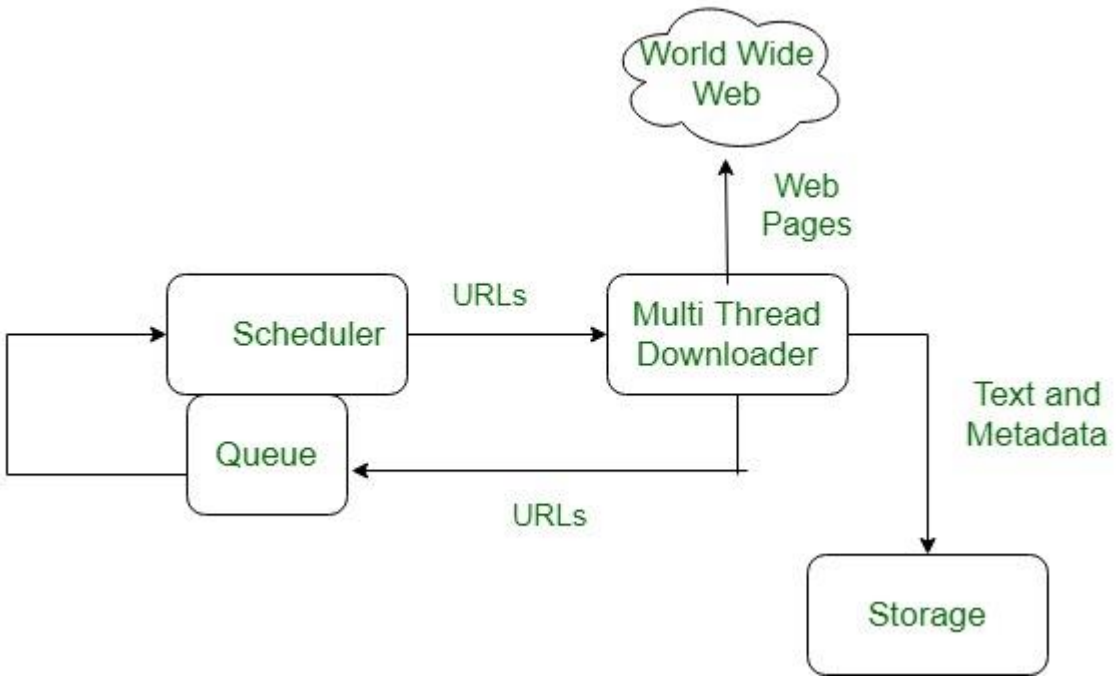
The **World Wide Web** is abbreviated as WWW and is commonly known as the web. The WWW was initiated by CERN (European library for Nuclear Research) in 1989.

System Architecture:

From the user's point of view, the web consists of a vast, worldwide connection of documents or web pages. Each page may contain links to other pages anywhere in the world. The pages can be retrieved and viewed by using browsers of which internet explorer, Netscape Navigator, Google Chrome, etc are the popular ones. The browser fetches the page requested interprets the text and formatting commands on it, and displays the page, properly formatted, on the screen.

The basic model of how the web works are shown in the figure below. Here the browser is displaying a web page on the client machine. When the user clicks on a line of text that is

linked to a page on the abd.com server, the browser follows the hyperlink by sending a message to the abd.com server asking it for the page.



Here the browser displays a web page on the client machine when the user clicks on a line of text that is linked to a page on abd.com, the browser follows the hyperlink by sending a message to the abd.com server asking for the page.

Working of WWW:

The World Wide Web is based on several different technologies: Web browsers, Hypertext Markup Language (HTML) and Hypertext Transfer Protocol (HTTP).

A Web browser is used to access web pages. Web browsers can be defined as programs which display text, data, pictures, animation and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers. Initially, Web browsers were used only for surfing the Web but now they have become more universal. Web browsers can be used for several tasks including conducting searches, mailing, transferring files, and much more. Some of the commonly used browsers are Internet Explorer, Opera Mini, and Google Chrome.

Features of WWW:

- HyperText Information System
- Cross-Platform

- Distributed
- Open Standards and Open Source
- Uses Web Browsers to provide a single interface for many services
- Dynamic, Interactive and Evolving.
- “Web 2.0”

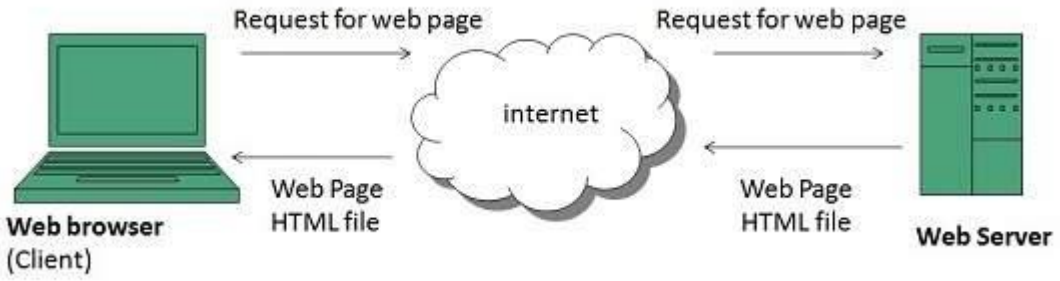
Components of the Web: There are 3 components of the web:

1. **Uniform Resource Locator (URL):** serves as a system for resources on the web.
2. **HyperText Transfer Protocol (HTTP):** specifies communication of browser and server.
3. **Hyper Text Markup Language (HTML):** defines the structure, organisation and content of a webpage.

WWW Operation

WWW works on client- server approach. Following steps explains how the web works:

1. User enters the URL (say, **http://www.tutorialspoint.com**) of the web page in the address bar of web browser.
2. Then browser requests the Domain Name Server for the IP address corresponding to www.tutorialspoint.com.
3. After receiving IP address, browser sends the request for web page to the web server using HTTP protocol which specifies the way the browser and web server communicates.
4. Then web server receives request using HTTP protocol and checks its search for the requested web page. If found it returns it back to the web browser and close the HTTP connection.
5. Now the web browser receives the web page, It interprets it and display the contents of web page in web browser’s window.



Dynamic Host Configuration Protocol (DHCP)

Prerequisite – Protocols in Application Layer

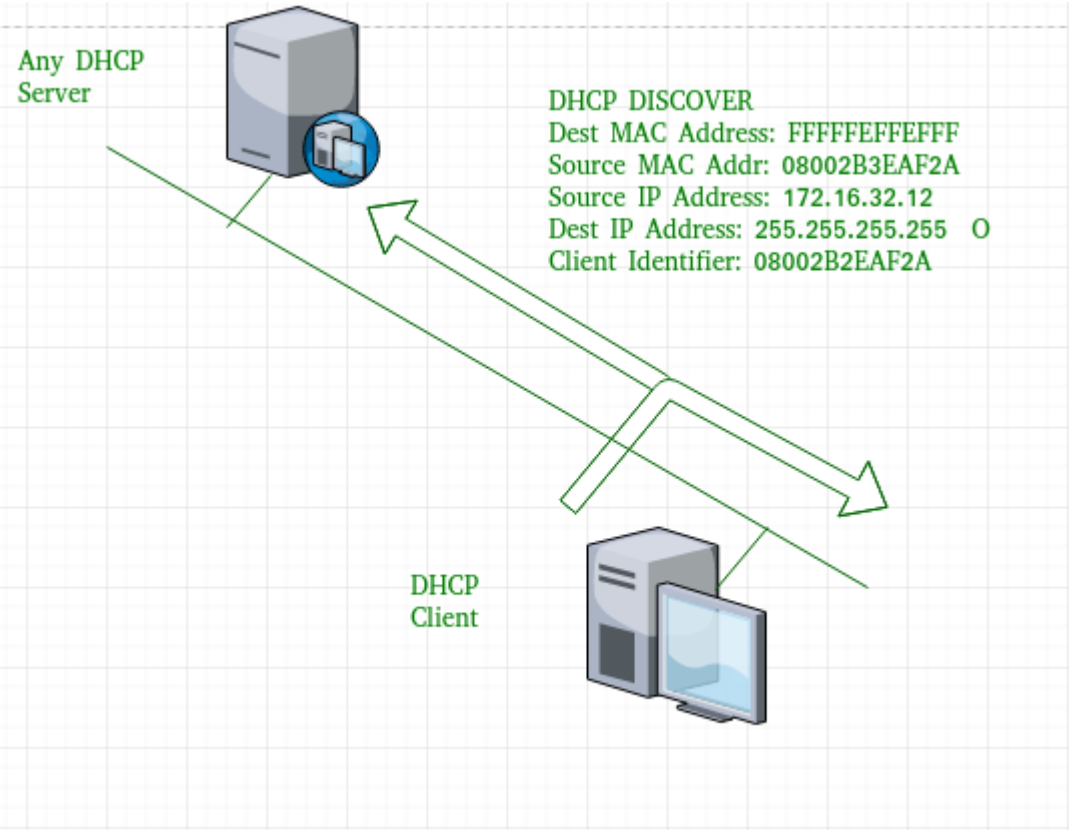
Dynamic Host Configuration Protocol(DHCP) is an application layer protocol which is used to provide:

1. Subnet Mask (Option 1 – e.g., 255.255.255.0)
2. Router Address (Option 3 – e.g., 192.168.1.1)
3. DNS Address (Option 6 – e.g., 8.8.8.8)
4. Vendor Class Identifier (Option 43 – e.g., ‘unifi’ = 192.168.1.9 ##where unifi = controller)

DHCP is based on a client-server model and based on discovery, offer, request, and ACK.

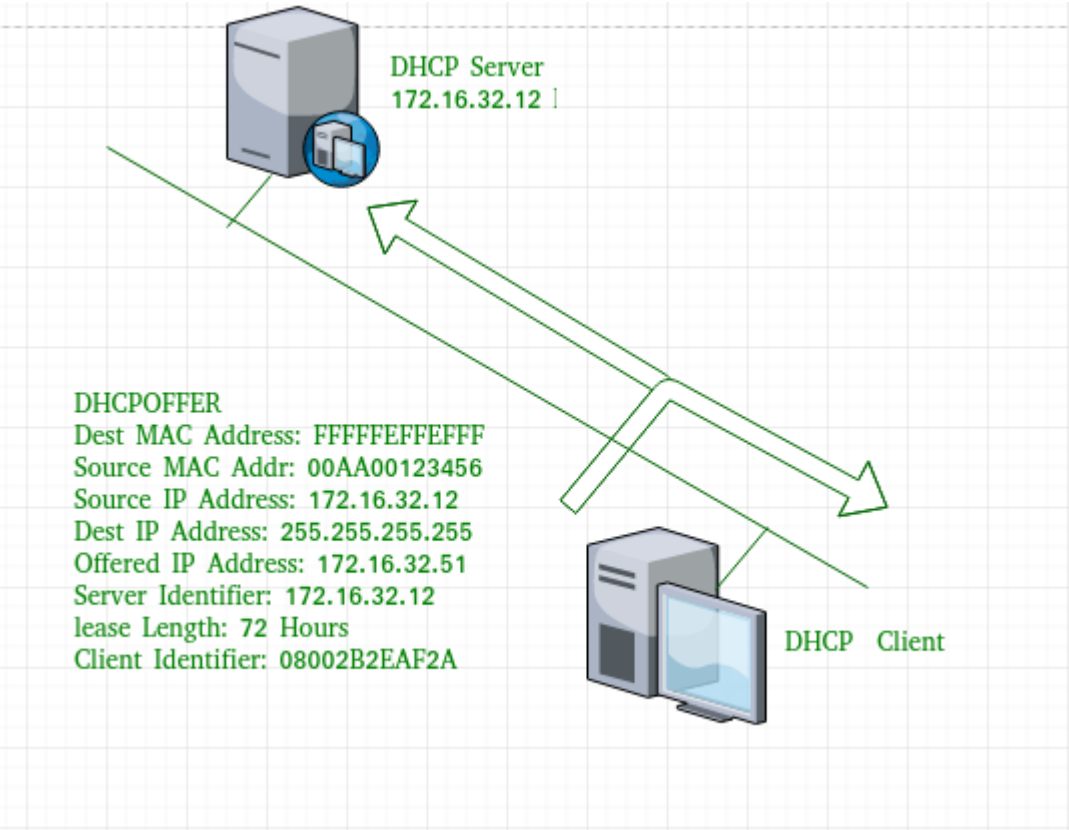
DHCP **port number** for server is 67 and for the client is 68. It is a Client server protocol which uses UDP services. IP address is assigned from a pool of addresses. In DHCP, the client and the server exchange mainly 4 DHCP messages in order to make a connection, also called DORA process, but there are 8 DHCP messages in the process.
These messages are given as below:

1. **DHCP discover message –**
   This is a first message generated in the communication process between server and client. This message is generated by Client host in order to discover if there is any DHCP server/servers are present in a network or not. This message is broadcasted to all devices present in a network to find the DHCP server. This message is 342 or 576 bytes long



1. As shown in the figure, source MAC address (client PC) is 08002B2EAF2A, destination MAC address(server) is FFFFFFFFFFFF, source IP address is 0.0.0.0(because PC has no IP address till now) and destination IP address is 255.255.255.255 (IP address used for broadcasting). As the discover message is broadcast to find out the DHCP server or servers in the network therefore broadcast IP address and MAC address is used.

2. **DHCP offer message –**
   The server will respond to host in this message specifying the unleased IP address and other TCP configuration information. This message is broadcasted by server. Size of message is 342 bytes. If there are more than one DHCP servers present in the network then client host will accept the first DHCP OFFER message it receives. Also a server ID is specified in the packet in order to identify the server.

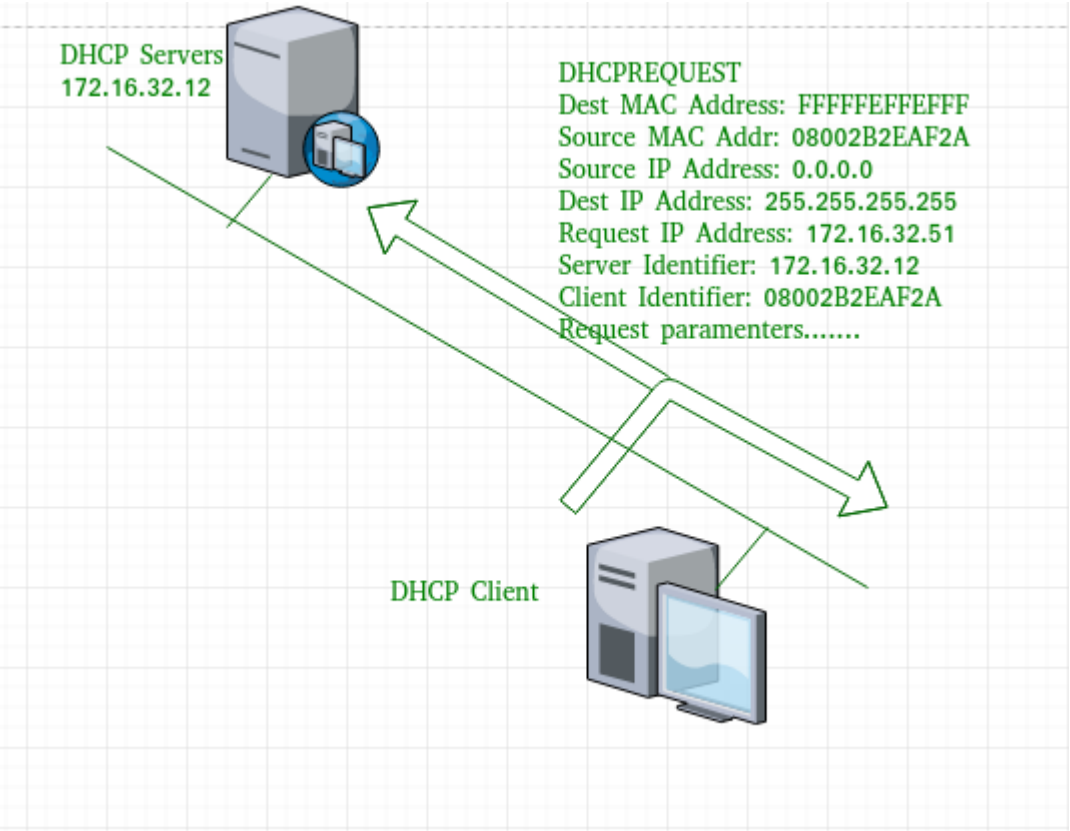


1. Now, for the offer message, source IP address is 172.16.32.12 (server's IP address in the example), destination IP address is 255.255.255.255 (broadcast IP address) ,source MAC address is 00AA00123456, destination MAC address is FFFFFFFFFFFF. Here, the offer message is broadcast by the DHCP server therefore destination IP address is broadcast IP address and destination MAC address is FFFFFFFFFFFF and the source IP address is server IP address and MAC address is server MAC address.

   Also the server has provided the offered IP address 192.16.32.51 and lease time of 72 hours(after this time the entry of host will be erased from the server automatically) . Also the client identifier is PC MAC address (08002B2EAF2A) for all the messages.

2. **DHCP request message –**

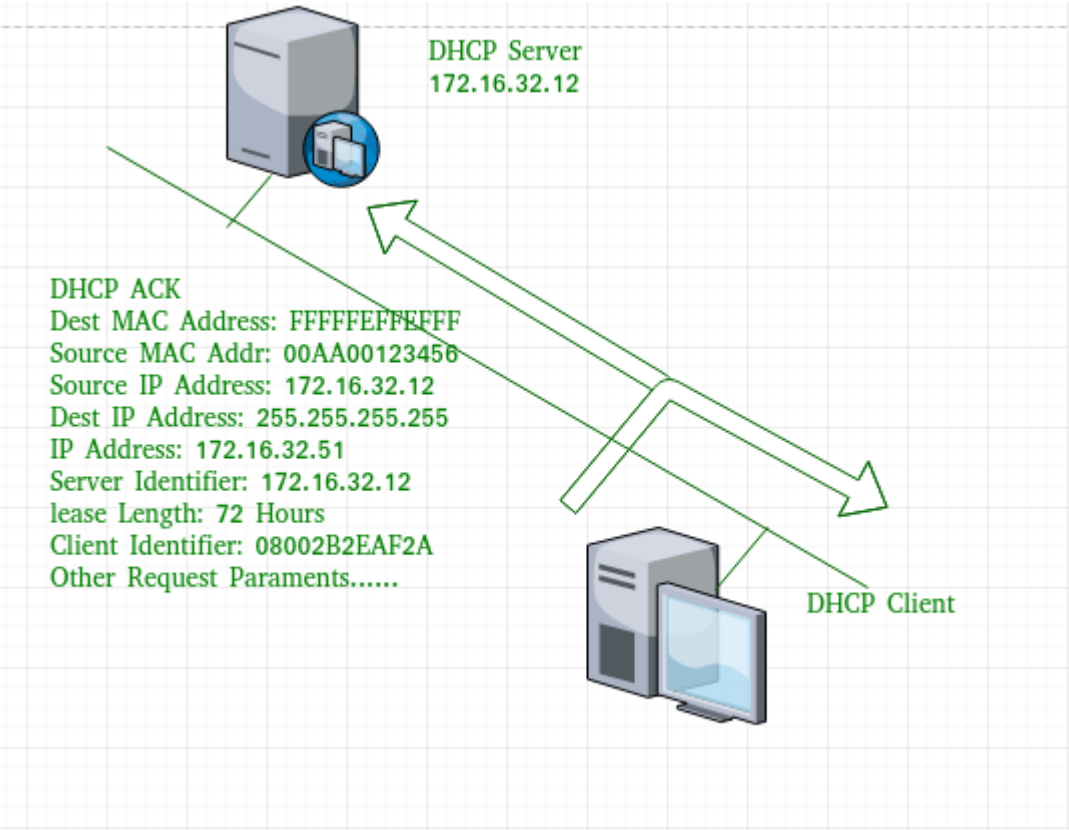
   When a client receives a offer message, it responds by broadcasting a DHCP request message. The client will produce a gratuitous ARP in order to find if there is any other host present in the network with same IP address. If there is no reply by other host, then there is no host with same TCP configuration in the network and the message is broadcasted to server showing the acceptance of IP address .A Client ID is also added in this message.



1. Now, the request message is broadcast by the client PC therefore source IP address is 0.0.0.0(as the client has no IP right now) and destination IP address is 255.255.255.255 (broadcast IP address) and source MAC address is 08002B2EAF2A (PC MAC address) and destination MAC address is FFFFFFFFFFFF.

   Note – This message is broadcast after the ARP request broadcast by the PC to find out whether any other host is not using that offered IP. If there is no reply, then the client host broadcast the DHCP request message for the server showing the acceptance of IP address and Other TCP/IP Configuration.

2. **DHCP acknowledgement message –**
   In response to the request message received, the server will make an entry with specified client ID and bind the IP address offered with lease time. Now, the client will have the IP address provided by server.



1. Now the server will make an entry of the client host with the offered IP address and lease time. This IP address will not be provided by server to any other host. The destination MAC address is FFFFFFFFFFFF and the destination IP address is 255.255.255.255 and the source IP address is 172.16.32.12 and the source MAC address is 00AA00123456 (server MAC address).

2. **DHCP negative acknowledgement message –**
   Whenever a DHCP server receives a request for IP address that is invalid according to the scopes that is configured with, it send DHCP Nak message to client. Eg-when the server has no IP address unused or the pool is empty, then this message is sent by the server to client.

3. **DHCP decline –**
   If DHCP client determines the offered configuration parameters are different or invalid, it sends DHCP decline message to the server .When there is a reply to the gratuitous ARP by any host to the client, the client sends DHCP decline message to the server

showing the offered IP address is already in use.

4. **DHCP release –**
   A DHCP client sends DHCP release packet to server to release IP address and cancel any remaining lease time.

5. **DHCP inform –**
   If a client address has obtained IP address manually then the client uses a DHCP inform to obtain other local configuration parameters, such as domain name. In reply to the dhcp inform message, DHCP server generates DHCP ack message with local configuration suitable for the client without allocating a new IP address. This DHCP ack message is unicast to the client.

Note – All the messages can be unicast also by dhcp relay agent if the server is present in different network.

Advantages – The advantages of using DHCP include:

- centralized management of IP addresses
- ease of adding new clients to a network
- reuse of IP addresses reducing the total number of IP addresses that are required
- simple reconfiguration of the IP address space on the DHCP server without needing to reconfigure each client

The DHCP protocol gives the network administrator a method to configure the network from a centralised area.

With the help of DHCP, easy handling of new users and reuse of IP address can be achieved.

Disadvantages – Disadvantage of using DHCP is:

- IP conflict can occur